Mehmet Kara

kara.28@osu.edu — 513-430-7631 — linkedin.com/in/karame

Education

The Ohio State University Columbus, OH
B.S. Mechanical Engineering May 2028

Technical Skills

Software & Simulation: CATIA V5, 3DEXPERIENCE, MATLAB, Simulink, Python, VBA, CarSim, ANSYS (FEA), Siemens TIA, SolidWorks, Siemens NX, Onshape, Autodesk Inventor, LabVIEW, Excel, Git
Manufacturing: CNC Milling, Manual Turning and Milling, Additive, Laser Cutting, MIG Welding
Languages: English (Native), Turkish (Fluent), Chinese (Intermediate)

Experience

Tesla Sparks, NV
Semi Manufacturing Engineering Intern May 2026 – August 2026

- Saved thousands in contracting costs by reverse engineering proprietary tools in CATIA V5 with laser scanning and CMM data, bringing modifications in-house.
- Designed and built a scaled down 6-axis Fanuc robot cell to train technicians on programming workflows, mirroring full scale production processes in a hands-on format.
- Reduced 5th wheel mounting station cycle time by 15% with a custom pneumatic quick connector, optimizing gasket sealing via design tables and SLA resin printing for a $5 in-house unit cost.

Formula Buckeyes (FSAE) Columbus, OH
Vehicle Dynamics Director August 2025 – Present

- Directed the mechanical redesign of a 600V, 9kWh battery pack for an electric formula race car, applying DFM principles in SolidWorks to deliver the team's first pack that could be fully assembled.
- Led 2026 and 2027 suspension design, building a parametric CAD and kinematic solver pipeline that automatically updated geometry and kinematics when hardpoints changed, cutting iteration time 75% and delivering kinematics early enough to set a realistic manufacturing schedule.
- Built the team's first battery testing and simulation program, validating temperature and voltage models built in Simulink against industry standard test methods to under 3% average residual, giving the team the ability to simulate future design changes.
- Designed the rear anti-roll bar to a 35% rear roll stiffness distribution target, using CarSim to set stiffness goals and an ANSYS FEA sweep to optimize material and wall thickness, landing within 0.5% while cutting mass versus the prior design.

Control Systems Lead August 2025 – July 2026

- Led development of four core control systems in Simulink including traction control and torque vectoring using Model-Based Design to bring the vehicle from open loop to fully closed loop control.

The Center for Automotive Research Columbus, OH
Vehicle Dynamics Research Assistant May 2025 – August 2025

- Led the creation of a custom Driver-in-the-Loop (DiL) vehicle simulator, allowing engineers to drive and test development vehicles in a virtual environment, significantly reducing development time.
- Validated simulator fidelity by correlating real-world and virtual results across industry-standard maneuvers, such as step steer tests, achieving a 98% correlation in lateral acceleration and yaw rate.
- Integrated bespoke plant models and control logic using Simulink, streamlining the transition from desktop simulation to real-time driver testing for processes like suspension and control systems design.
- Collaborated with industry leaders, Applied Intuition, to develop a low-latency data link for driver peripherals.

Bearcat Motorsports (FSAE) Cincinnati, OH
Suspension Team Member September 2023 – April 2024

- Designed a carbon fiber composite steering wheel in Siemens NX and ANSYS ACP, achieving a 20% mass reduction over the outgoing metallic design.